Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - III

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PART – A: Theory

Python provides built-in modules to work with **date**, **time**, and **calendar**. These modules help programmers display the current date, convert timestamps into readable dates, generate calendars, and compare dates. They are widely used in attendance systems, scheduling applications, event management, and report generation.

1. Working with Date and Time (`datetime` Module)

The `datetime` module provides classes to work with dates and time. It allows us to get the current date and time, format them, and perform calculations.

Syntax:

```
from datetime import datetime
today = datetime.now()
```

Explanation:

- `from datetime import datetime` imports the `datetime` class.
- `datetime.now()` returns the current system date and time.

2. Working with Timestamps (`time` Module)

A timestamp is the number of seconds elapsed since **January 1, 1970 (Unix Epoch)**. The `time` module converts timestamps into a readable date and time.

Syntax:

```
import time
time.ctime(timestamp)
```

Explanation:

- `ctime()` converts a timestamp into a human-readable date and time.

3. Displaying a Calendar (`calendar` Module)

The `calendar` module is used to display monthly and yearly calendars in Python.

Syntax:

```
import calendar
calendar.month(year, month)
```

Explanation:

- `month()` displays the calendar for a given month and year.
- `calendar()` displays the complete calendar for a year.

4. Comparing Two Dates

Python allows comparison of dates using comparison operators or by calculating the difference between dates.

Syntax:

```
date1 > date2
date2 - date1
```

Explanation:

- `>` checks which date is later.
- `-` returns the difference between two dates.

PART – B Questions:

a. Write a python code to print current date in different format.

CODE:

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File Edit Format Run Options Window Help

```
from datetime import datetime

d = datetime.now()
print(d.strftime("%d-%m-%Y"))
print(d.strftime("%d-%m-%y"))
print(d.strftime("%B-%d-%Y"))
```

OUTPUT:

```
>>>
==================
29-07-2026
29-07-26
July-29-2026
>>>
```

b. Write a python code to convert time stamp to date stamp.

CODE:

```python
import time
t=time.time()
print(t)
print(time.ctime(t))
```

OUTPUT:

```
>>
================================
1785313952.962028
Wed Jul 29 14:02:32 2026
>>
```

c. Write a python code to develop calendar module.

CODE:

```python
import calendar
print(calendar.month(2007,3))
```

OUTPUT:

```
===========================
     March 2007
Mo Tu We Th Fr Sa Su
          1  2  3  4
 5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31
```

d. Write a python code to compare two dates.

CODE:

```
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from datetime import date

d1=date(2026,7,29)
d2=date(2026,9,14)

print(d1>d2)
print(d2-d1)
```

OUTPUT:

```
========================
False
47 days, 0:00:00
```

Curiosity Question:

1. **Why do computers use timestamps instead of storing dates directly?**
2. **How does Python know the current date and time?**
3. **Can Python display the calendar for any year, even 100 years in the future?**
4. **Why is comparing dates important in real-world applications?**
5. **What happens if you compare two dates from different years?**